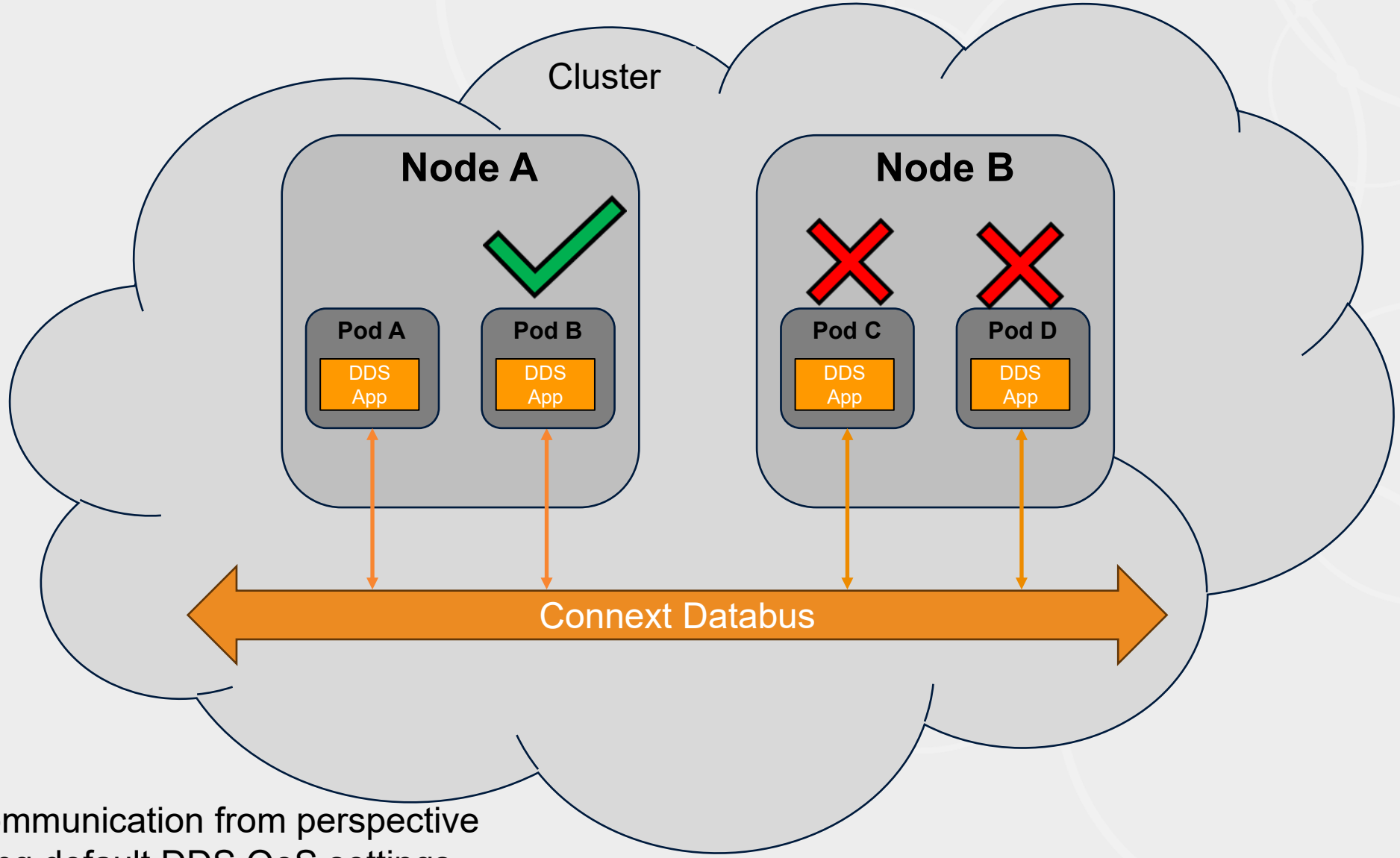
Kubernetes Deployments

Flannel CNI

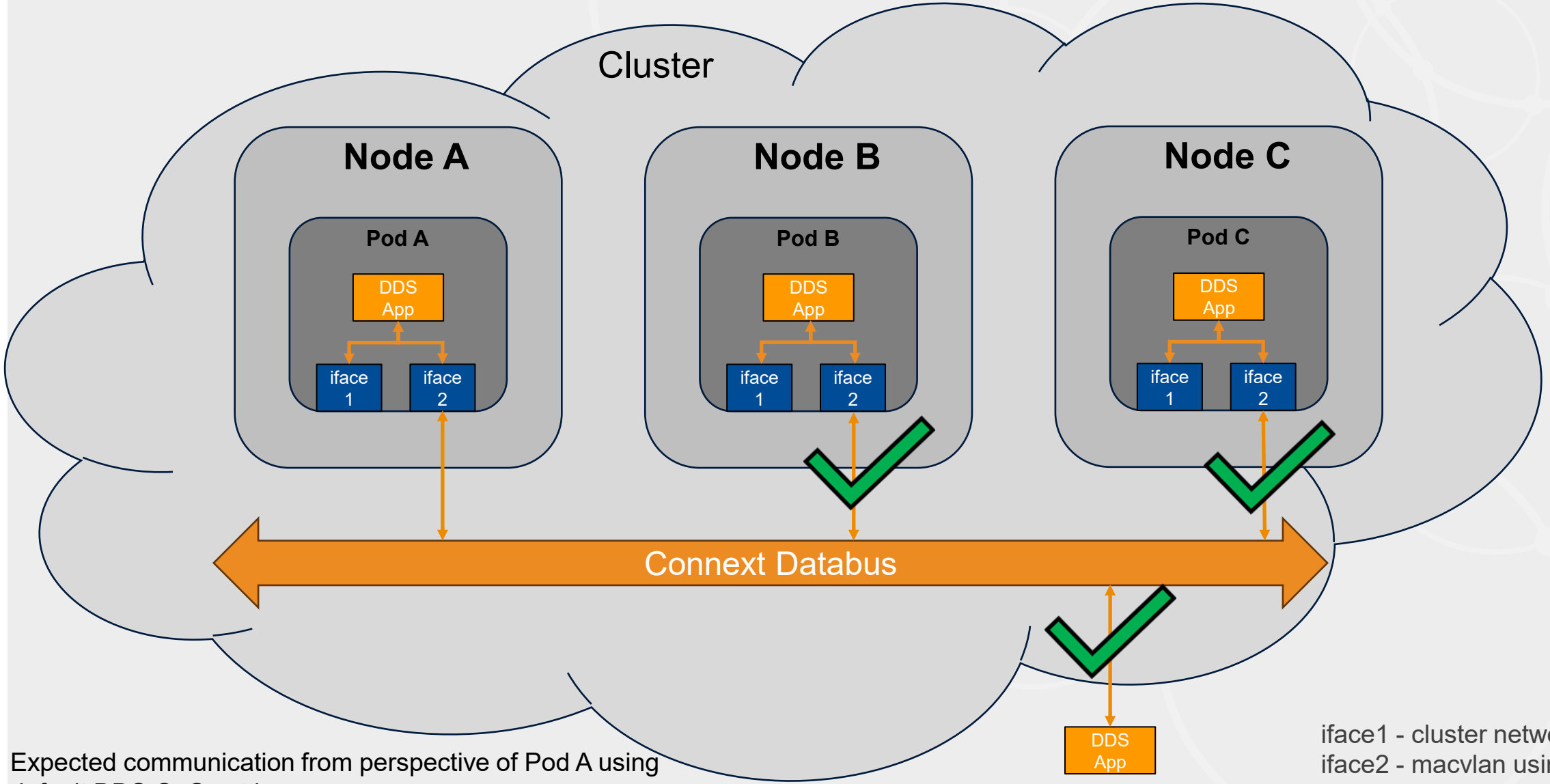
DDS Communication with Flannel



Expected communication from perspective of Pod A using default DDS QoS settings

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DDS Communication with Flannel + Multus

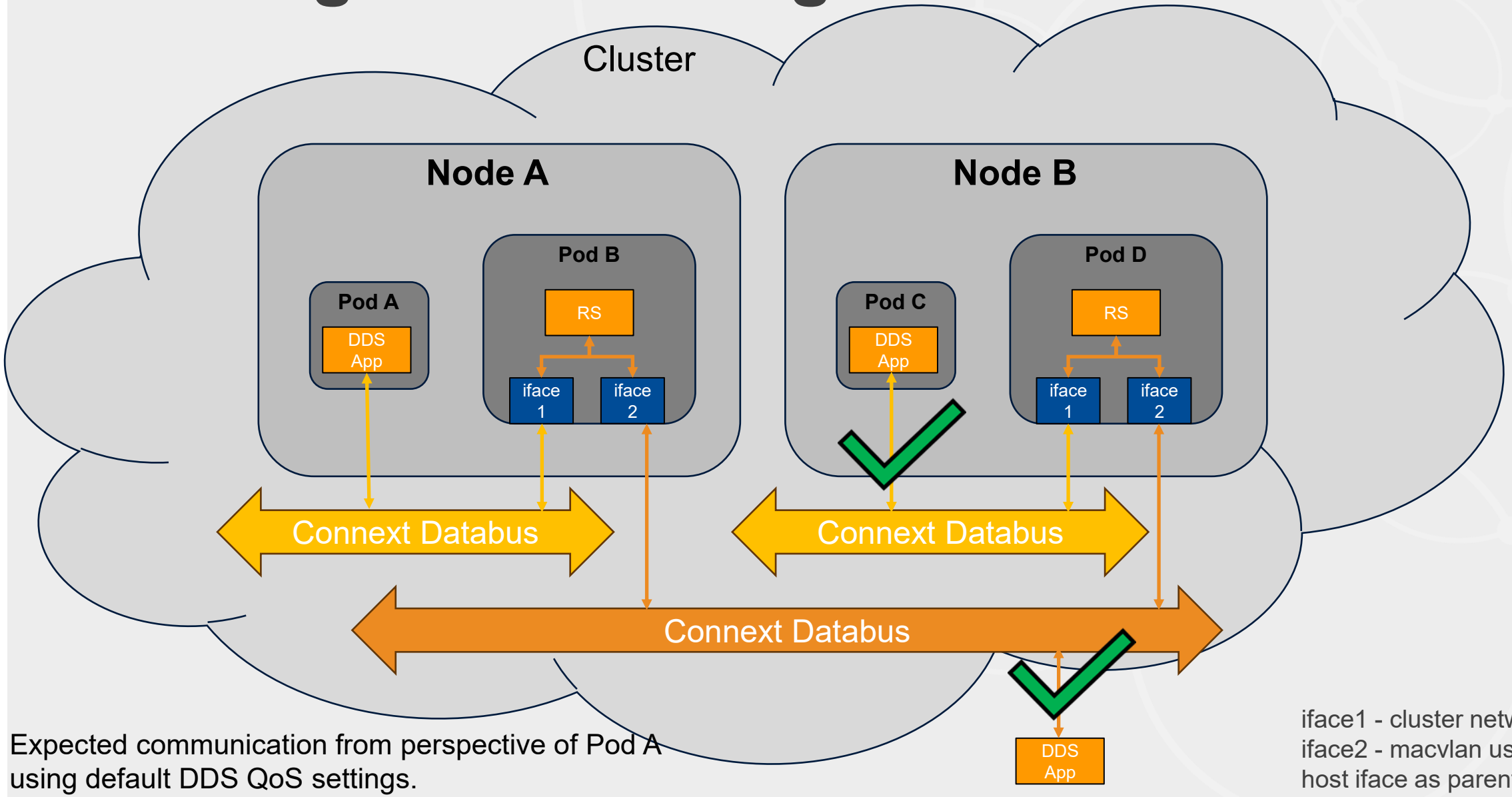


Expected communication from perspective of Pod A using default DDS QoS settings.

iface1 - cluster network
iface2 - macvlan using host iface as parent

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Routing Service Bridge – Flannel + Multus

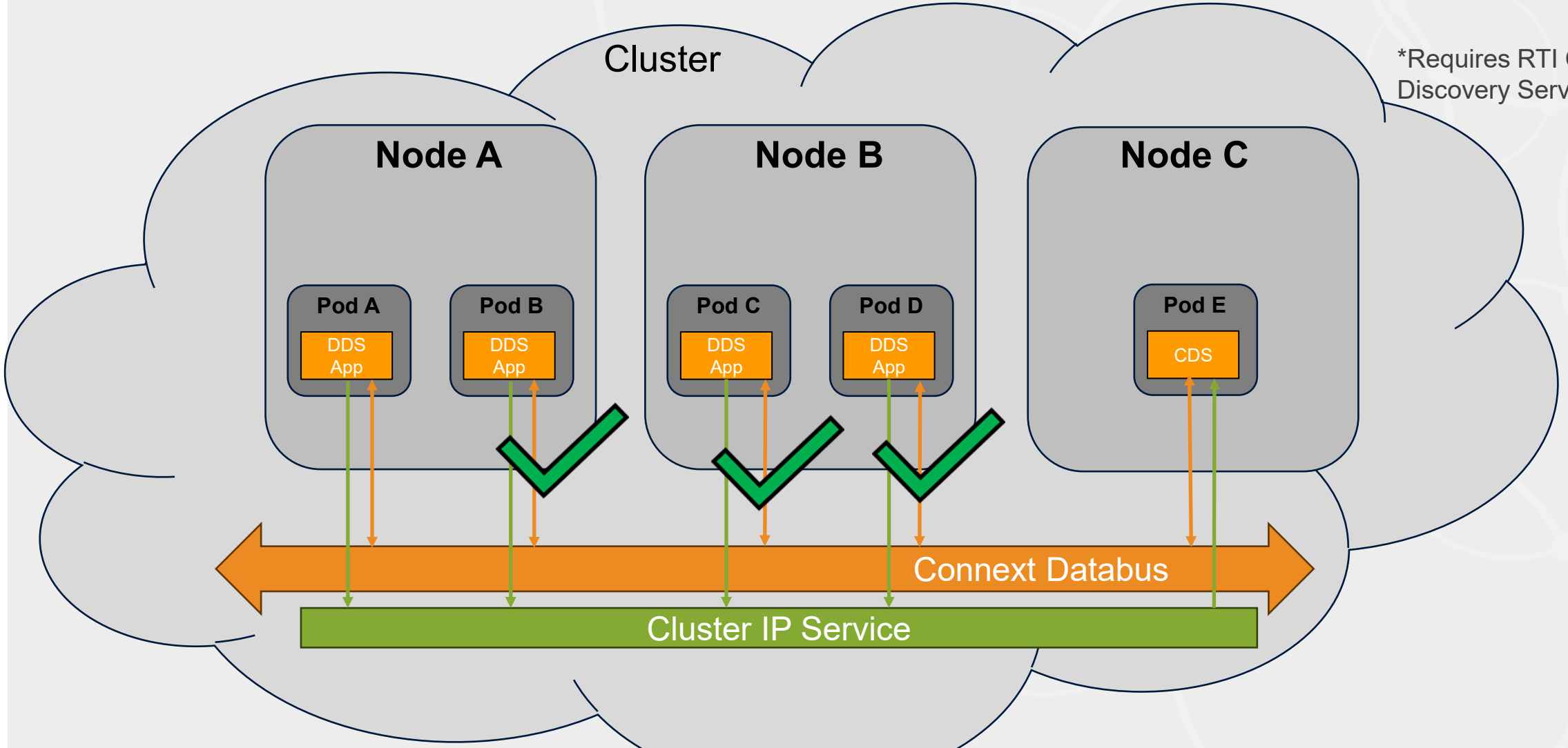


Expected communication from perspective of Pod A using default DDS QoS settings.

iface1 - cluster network
iface2 - macvlan using host iface as parent

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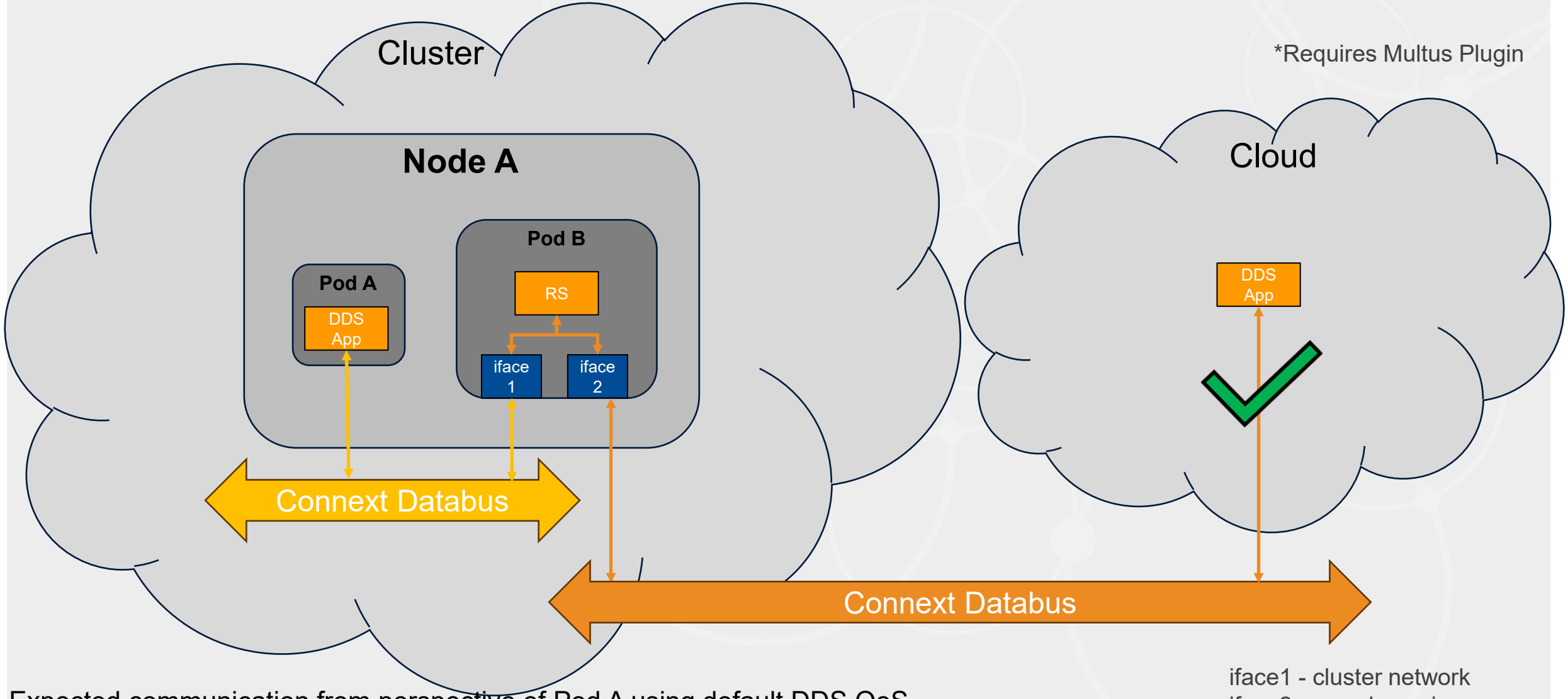
DDS Communication without Multicast*



Cluster IP Service exposes static IP/Port within the cluster. DDS apps use static IP/Port in initial peers list. CDS enables discovery.

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DDS Communication to Cloud*



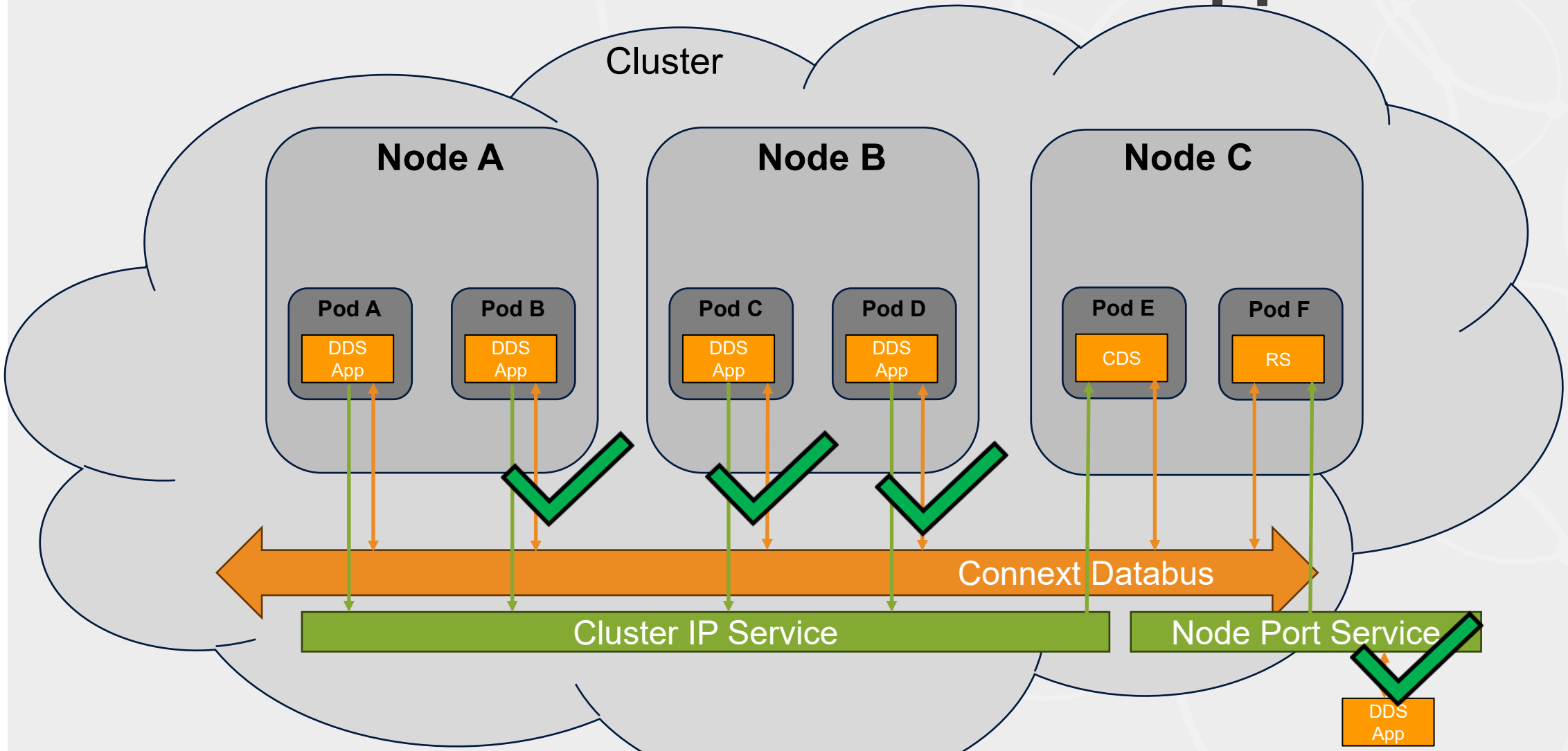
*Requires Multus Plugin

Expected communication from perspective of Pod A using default DDS QoS settings. Routing Service bridge allows communication outside of the cluster.

iface1 - cluster network
iface2 - macvlan using host iface as parent

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DDS Communication External Applications



Cluster IP Service exposes static IP/Port within the cluster. DDS apps use static IP/Port in initial peers list. CDS enables discovery.

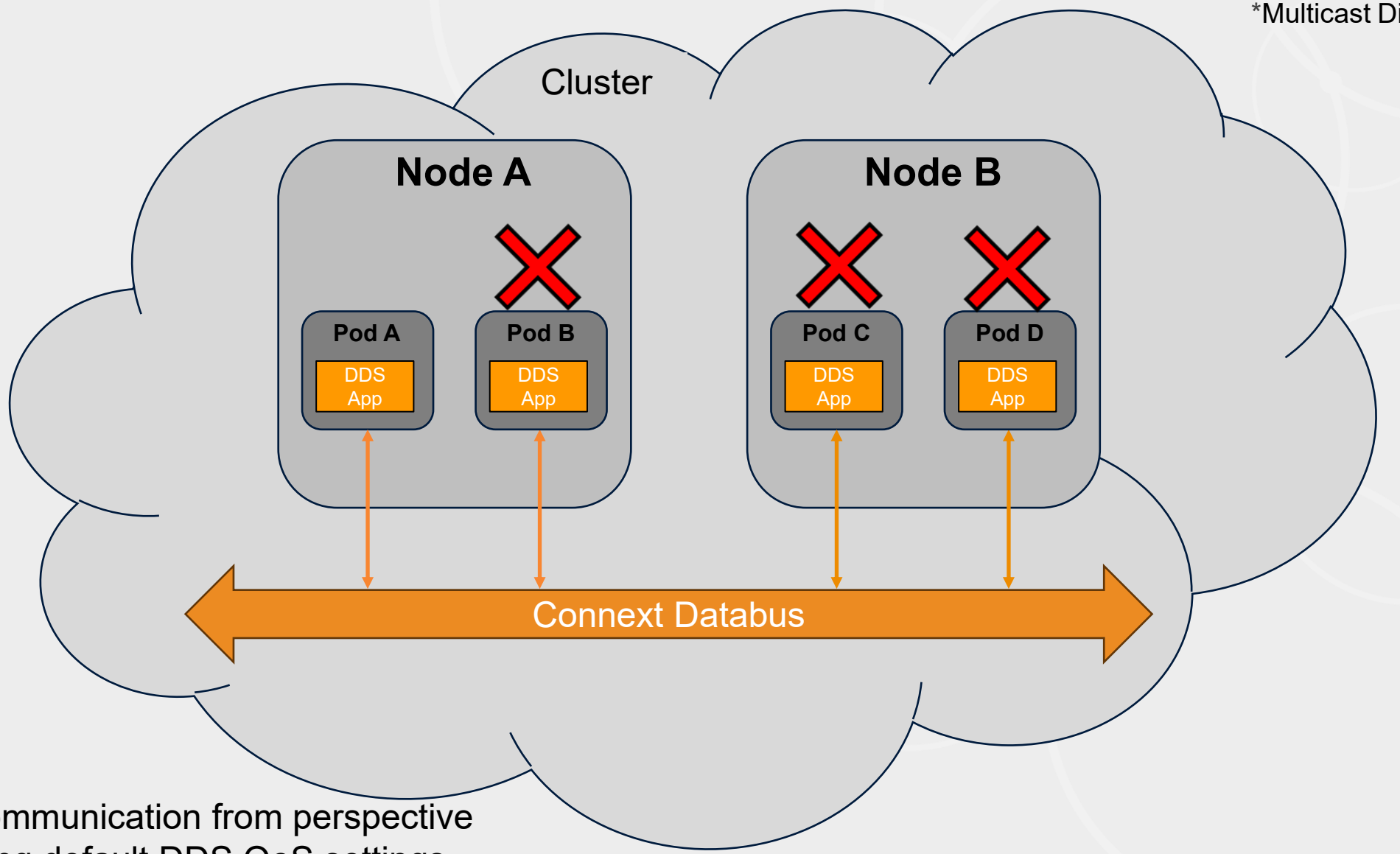
Node port Service exposes static IP/Port to outside of cluster. External DDS apps use static IP/Port in initial peers list.

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Antrea CNI

DDS Communication with Antrea

*Multicast Disabled

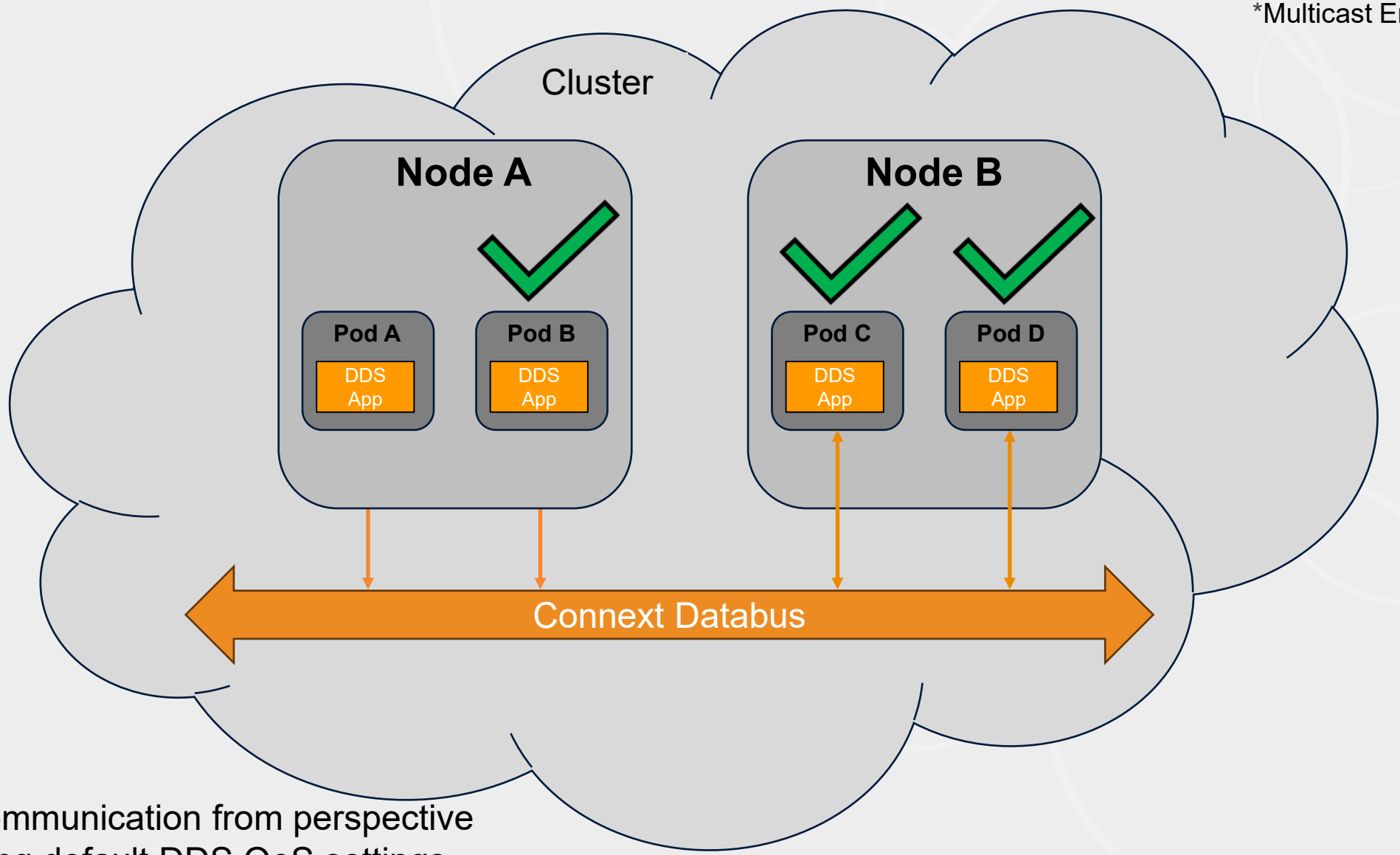


Expected communication from perspective of Pod A using default DDS QoS settings

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DDS Communication with Antrea*

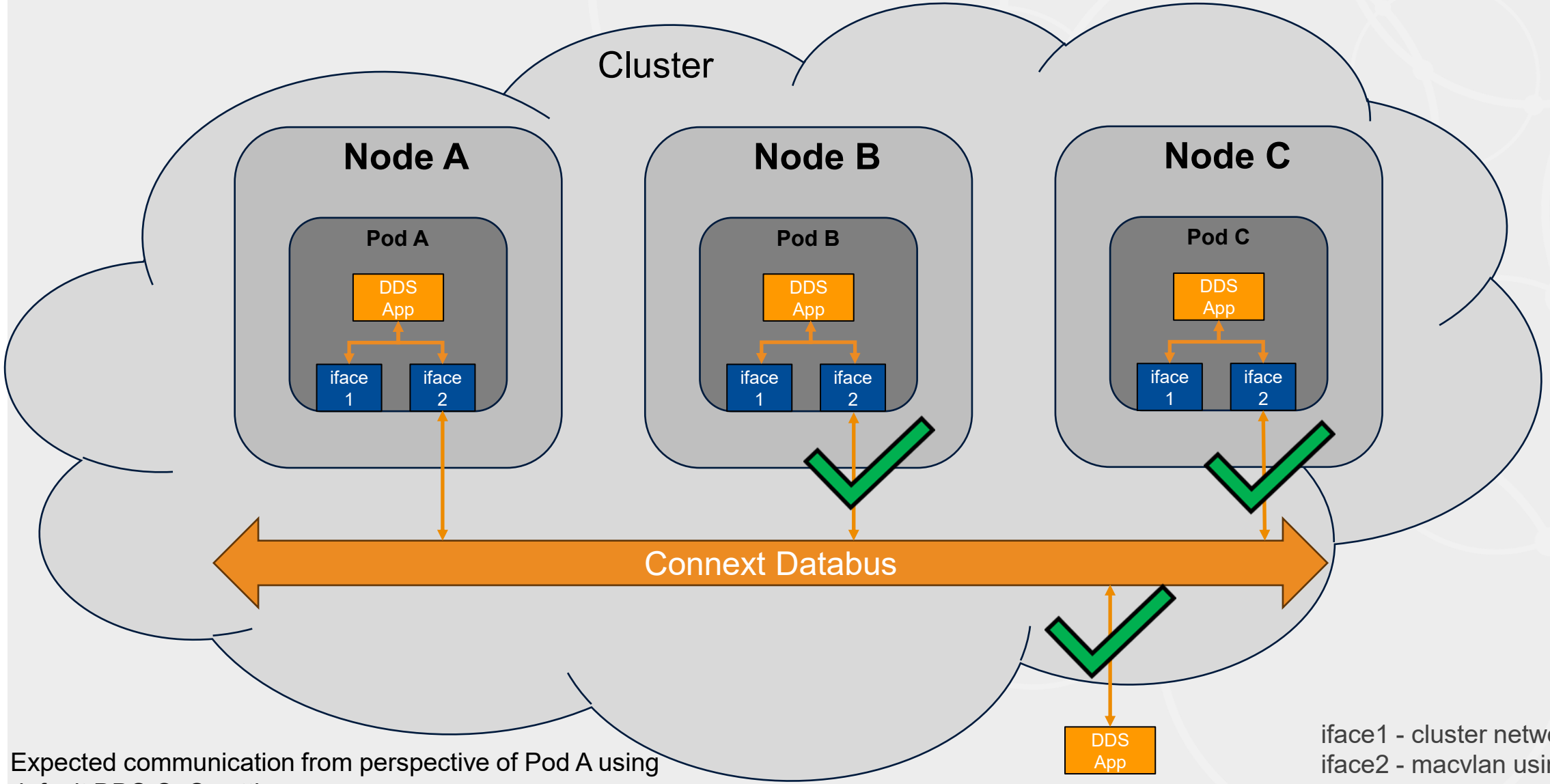
*Multicast Enabled



Expected communication from perspective of Pod A using default DDS QoS settings

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DDS Communication with Antrea + Multus

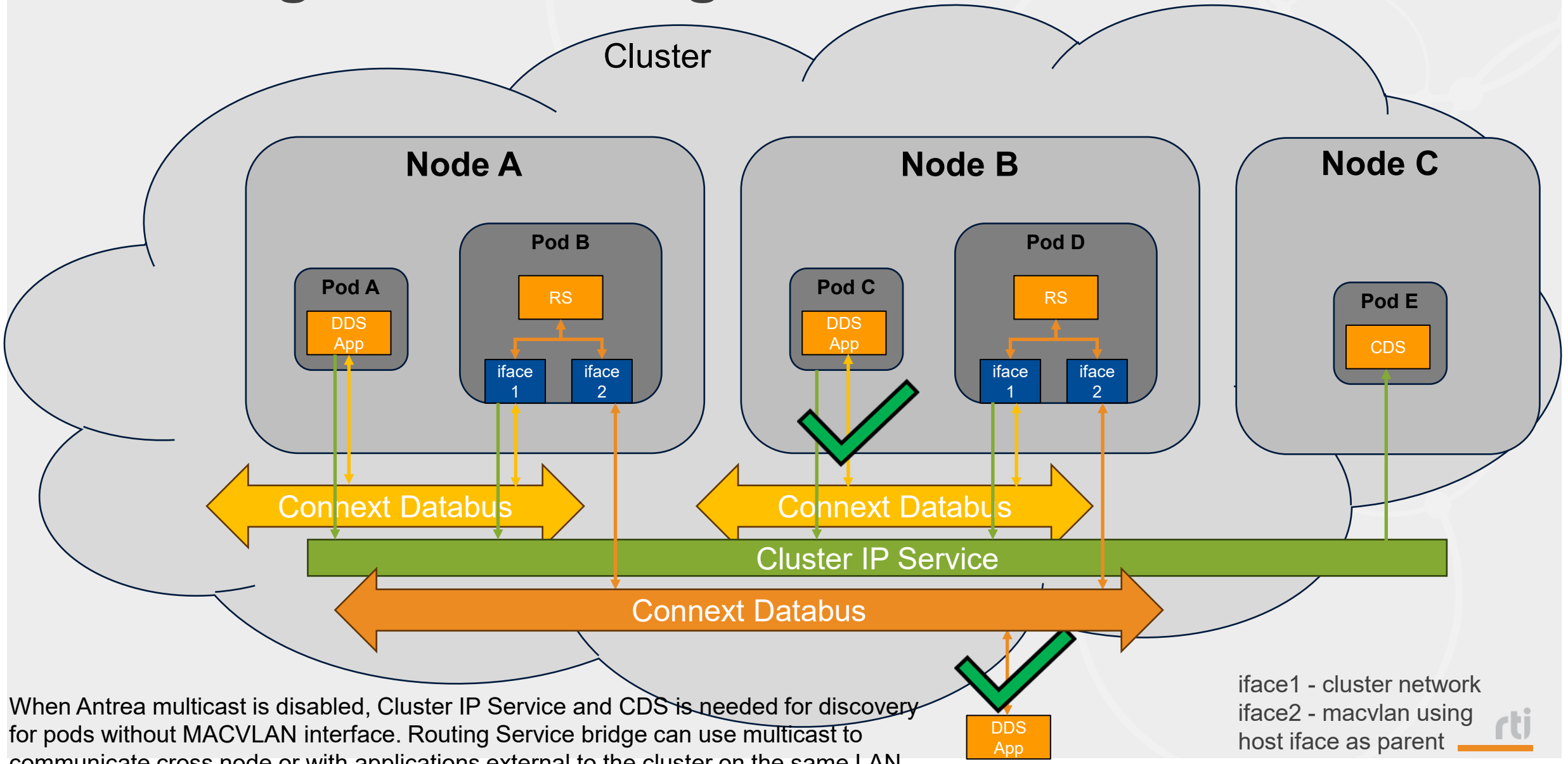


Expected communication from perspective of Pod A using default DDS QoS settings.

iface1 - cluster network
iface2 - macvlan using host iface as parent

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Routing Service Bridge – Antrea + Multus

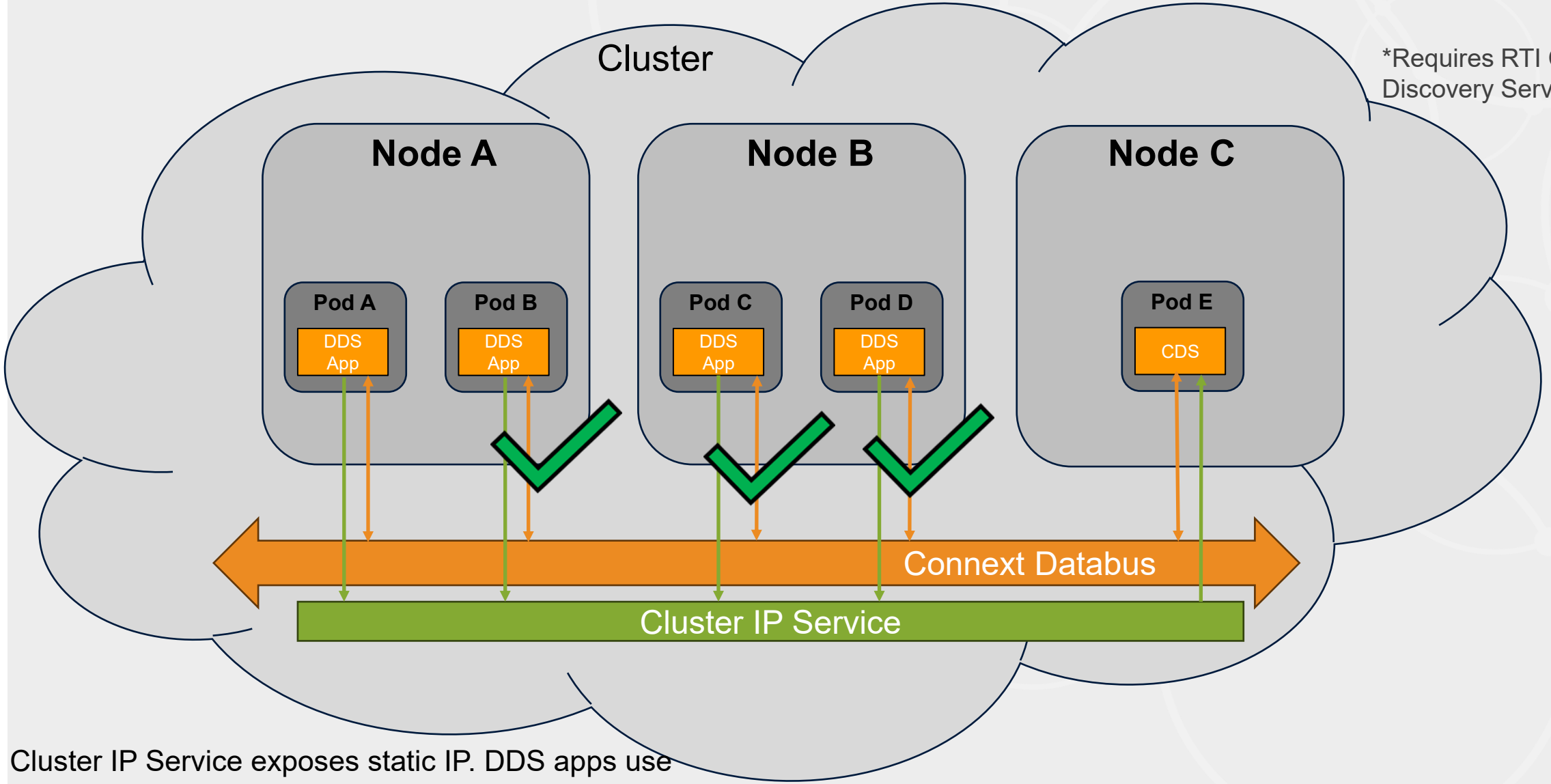


When Antrea multicast is disabled, Cluster IP Service and CDS is needed for discovery for pods without MACVLAN interface. Routing Service bridge can use multicast to communicate cross node or with applications external to the cluster on the same LAN.

iface1 - cluster network
iface2 - macvlan using host iface as parent

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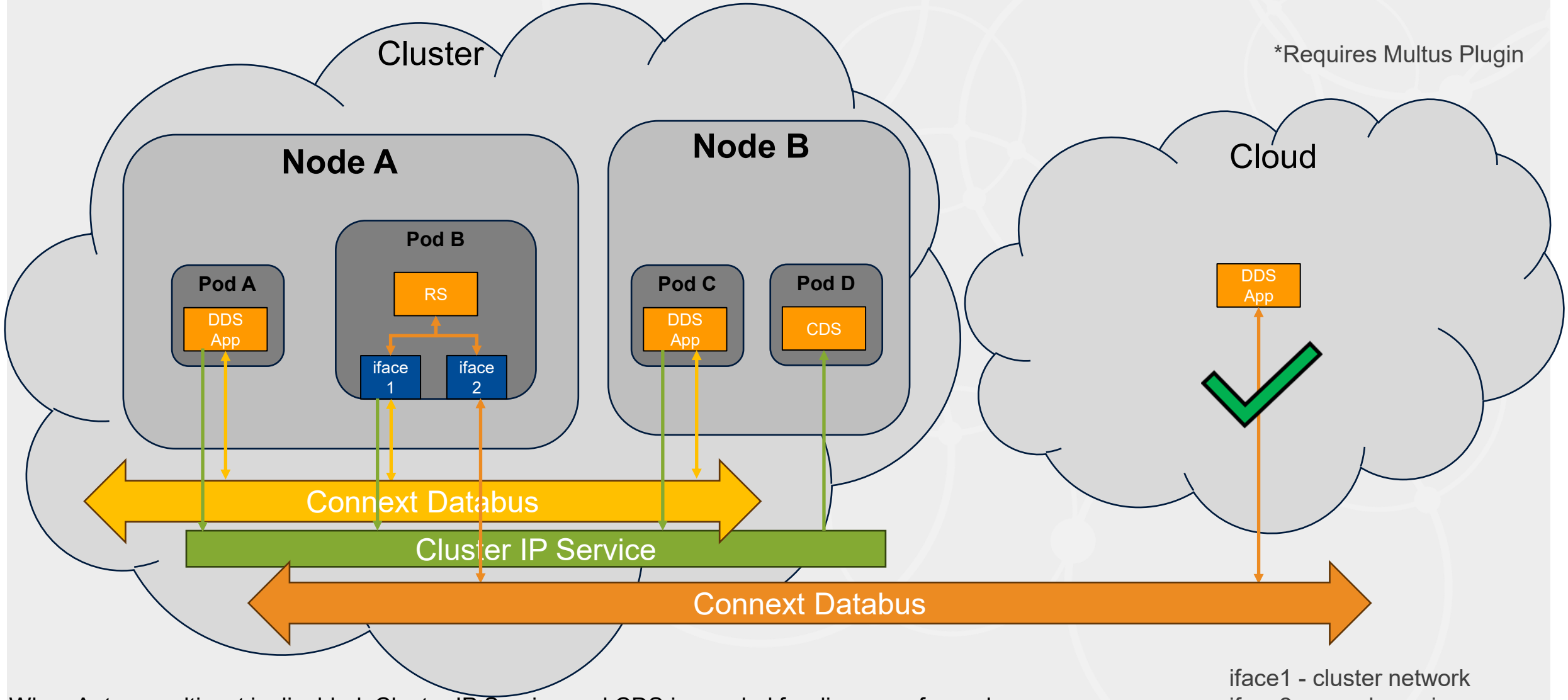
DDS Communication without Multicast*



Cluster IP Service exposes static IP. DDS apps use static IP in initial peers list. CDS enables discovery.

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DDS Communication to Cloud*



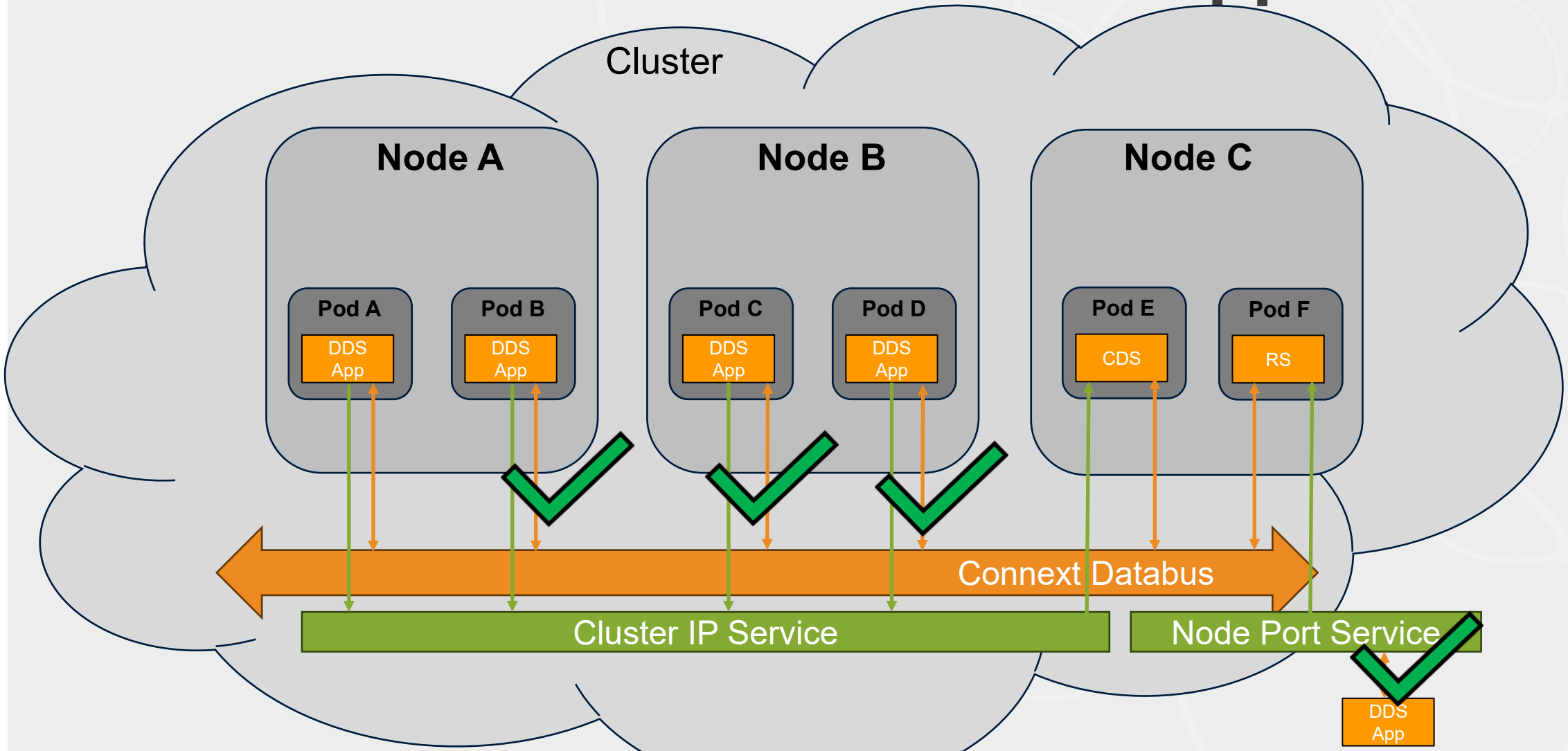
*Requires Multus Plugin

When Antrea multicast is disabled, Cluster IP Service and CDS is needed for discovery for pods without MACVLAN interface. Routing Service bridge allows communication outside of the cluster.

iface1 - cluster network
iface2 - macvlan using host iface as parent

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DDS Communication External Applications



Cluster IP Service exposes static IP/Port within the cluster. DDS apps use static IP/Port in initial peers list. CDS enables discovery.

Node port Service exposes static IP/Port to outside of cluster. External DDS apps use static IP/Port in initial peers list.

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